BEFORE YOU READ

The roots pages

The roots section splits into eight pages, from the basic properties of radicals to their use in equations, functions and the complex numbers.

- L0 **Roots and radicals**
  - L1 **Rules**
    - L2 **Properties of radicals** even vs odd index, sign, domain
    - L2 **Radical rules** product, quotient, power, nested
    - L2 **Rational exponents** fractional powers, converting forms
  - L1 **Working with radicals**
    - L2 **Simplifying radicals** extract perfect powers, clear fractions
    - L2 **Operations with radicals** add like radicals, rationalize
  - L1 **Applications**
    - L2 **Radical equations** isolate the radical; extraneous solutions
    - L2 **Radical functions** domains and graphs of $\sqrt{x}$ and $\sqrt[3]{x}$
    - L2 **Radicals and complex numbers** negative radicands through i

TAKEAWAY

Rules first, then use

Properties and rules come first; simplifying and operations put them to work; equations, functions and complex numbers apply them.